Number Spiral

A number spiral is an infinite grid whose upper-left square has number 1. Here are the first five layers of the spiral:

1	2	9	10	25
4	3	8	11	24
5	6	7	12	23
16	15	14	13	22
17	18	19	20	21

find out the number in row y and column x.

Input

The first input line contains an integer t: the number of tests.
After this, there are t lines, each containing integers y and x.

Example

Input: 2 3

Output: 8

Approach

The grid can be visualized as consisting of concentric squares.

The value at position (Y, X) always lies on the boundary of the square whose side length is the maximum of Y and X.

To compute this value, we first calculate the area of the inner square with side length (max(Y, X) - 1).

The remaining steps to reach the target position can then be added to this base value.

The direction in which the numbers increase or decrease depends on the parity of the side:

- If the side length is even, the numbers along the boundary are filled in decreasing order.

- If the side length is odd, the numbers along the boundary are filled in increasing order in the anticlockwise direction.

By using this observation, the value at (Y, X) can be determined directly without constructing the entire grid.